

MedSegMNIST: Accessible 2D and 3D Medical Image Segmentation Datasets for Research and Education

Toufiq Musah ¹ 

¹ Kwame Nkrumah University of Science and Technology, Kumasi, Ghana

Abstract

Automated medical image segmentation is a fundamental task in medical image analysis, supported by a growing number of publicly available datasets that span diverse imaging modalities and anatomies. Segmentation datasets remain highly fragmented across repositories and formats, creating barriers to reproducible benchmarking, educational use, and rapid prototyping. This work presents MedSegMNIST, an open-source benchmark and software package designed to standardize access to medical image segmentation datasets and streamline reproducible experimentation. MedSegMNIST currently integrates ten segmentation datasets spanning seven imaging modalities, multiple anatomical regions, and both binary and multi-class segmentation tasks in 2D and 3D. The resource aims to facilitate reproducible benchmarking, rapid prototyping, and exploratory learning across a diverse range of segmentation applications. Code and data available at: <https://github.com/toufiqmusah/MedSegMNIST>

Keywords

Medical image segmentation, Open datasets, Benchmarking, Deep Learning

Article informations

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Corresponding author: tmusah@st.knust.edu.gh

1. Background

Availability of publicly accessible medical imaging datasets has expanded considerably over the past decade (Jiménez-Sánchez et al., 2025; Alabduljabbar et al., 2024). This is primarily driven by growing interest in artificial intelligence applications in image analysis and the increasing adoption of open-science practices. Large collections of annotated images are now available across a diverse range of imaging modalities (Simpson et al., 2019; Clark et al., 2013; Sudlow et al., 2015), including magnetic resonance imaging (MRI), computed tomography (CT), ultrasound, radiography, pathology, fundus photography, dermoscopy, and endoscopy. They span several anatomical regions and clinical applications, providing a valuable resource for the development and evaluation of medical image analysis algorithms.

Despite wide availability, medical image segmentation datasets remain highly fragmented (Deng et al., 2026). Individual datasets are distributed through different repositories, stored in heterogeneous formats, and accompanied by varying preprocessing procedures and evaluation protocols. Access requirements may differ substantially between resources, ranging from direct downloads to registration-based systems or special transfer tools.

Table 1: Feature comparison against existing medical segmentation resources. full support (✓), limited support (~), not supported (×).

Feature	MSD	MedSegBench	MedSegMNIST
2D support	×	✓	✓
3D support	✓	×	✓
Unified API	×	✓	✓
CV Splits	×	×	✓
Native-resolution	✓	×	✓
Multi-class labels	✓	✓	✓
Benchmark pipeline	×	~	✓

This fragmentation presents challenges for exploratory learning and or reproducible benchmarking. Differences in preprocessing pipelines, train-test splits, image resolutions, and label representations can lead to inconsistencies across studies. These issues are particularly relevant for segmentation tasks. Further, many segmentation datasets are large and computationally demanding, limiting their accessibility for education (exploratory learning), and rapid prototyping.

2. Summary

To address these challenges, we present MedSegMNIST, an accessible source for multi-resolution 2D and 3D medical image segmentation datasets for education, research,

and benchmarking purposes. MedSegMNIST provides standardized access to ten segmentation datasets spanning seven imaging modalities and a broad range of anatomical structures. The resource includes both 2D and 3D segmentation tasks, multiple resolution variants including original native-resolution, predefined cross-validation splits, support for binary and multi-class segmentation, and a consistent Python interface designed for rapid experimentation and reproducible evaluation. The primary contributions of MedSegMNIST are:

- A curated collection of ten medical image segmentation datasets spanning multiple modalities, anatomies, and clinical applications.
- Access to multiple resolution 2D and 3D segmentation tasks, with predefined cross-validation splits through a common dataset interface to facilitate reproducible benchmarking.
- Integrated utilities that enable rapid exploratory learning, visualization, model development, and bring-your-own-model benchmarking with minimal setup requirements.

By reducing the technical overhead associated with dataset preparation and evaluation, MedSegMNIST aims to make medical image segmentation research more accessible while promoting reproducible benchmarking across a diverse set of segmentation tasks.

3. Related Works

The success of benchmark resources such as MNIST demonstrated the value of standardized, easy-to-use datasets for accelerating methodological development and reproducible evaluation. More recently, MedMNIST (Yang et al., 2023) extended this to medical image classification by providing lightweight, standardized datasets that lower the barrier to entry for medical imaging research. For medical image segmentation, initiatives such as MedSegBench (Kuş and Aydın, 2024) have similarly sought to facilitate benchmarking across multiple datasets. Existing resources primarily focus on curated benchmark collections and do not simultaneously provide lightweight access, native-resolution support, standardized cross-validation protocols, and a common interface supporting both two-dimensional (2D) and three-dimensional (3D) segmentation tasks.

4. Dataset Preparation

Source files (DICOM series, NIfTI volumes, multi-format 2D images) are converted to a common in-memory array representation. For volumetric datasets, images and corresponding segmentation masks are first reoriented to a

common anatomical axis convention before further processing. Processed datasets are distributed as compressed NPZ archives and are to be hosted on Zenodo with the following reserved DOI: 10.5281/zenodo.20694762, following the MedMNIST convention (Yang et al., 2023). This enables programmatic downloading of specific curated datasets of choice through the MedSegMNIST API.

Modality-aware normalisation CT volumes are clipped to a fixed Hounsfield Unit window ($[-1000, 1000]$) before foreground z-scoring; MRI and X-ray images are clipped at modality-specific intensity percentiles ($[0.5, 99.5]$ for MRI, $[1, 99]$ for X-ray) before z-scoring; RGB modalities (endoscopy, dermoscopy, fundus, histopathology) are scaled to $[0, 1]$.

Multi-resolution resampling Each volume or image is resampled to a small set of standardised target sizes using TorchIO (Pérez-García et al., 2021) for 3D data, with images resampled by trilinear interpolation and segmentation masks resampled by nearest-neighbour interpolation to preserve discrete label integrity. For 2D data bilinear and nearest-neighbour interpolations are applied to the image and mask respectively. A native-resolution variant is additionally produced for each dataset, preserving original per-image dimensions to support evaluation in physical millimetre units.

Split generation An 80/20 train/test partition and five-fold cross-validation index sets are computed once, with a fixed random seed, and stored as sample indices in a companion JSON file rather than being recomputed per training run, ensuring that every user of a given dataset trains and evaluates on identical partitions.

Packaging Final arrays are written as compressed NPZ files, following the Yang et al. (2023) convention, alongside a JSON metadata sidecar (containing class names, voxel spacing, license, source citation, split indices) and a SHA-256 checksum file for integrity verification on download.

5. MedSegMNIST Datasets

MedSegMNIST currently comprises ten segmentation datasets spanning seven imaging modalities and both two-dimensional (2D) and three-dimensional (3D) image analysis tasks. The collection was curated to provide diversity across anatomies, imaging characteristics, and segmentation objectives while maintaining a common interface for data access and evaluation. Included datasets cover organ, lesion, vessel, tissue, and tumour segmentation problems in CT, MRI, radiography, ultrasound, fundus photography, dermoscopy, endoscopy, and digital pathology. Together, these datasets expose users to a broad range of medical image segmentation settings, from binary boundary delineation tasks to

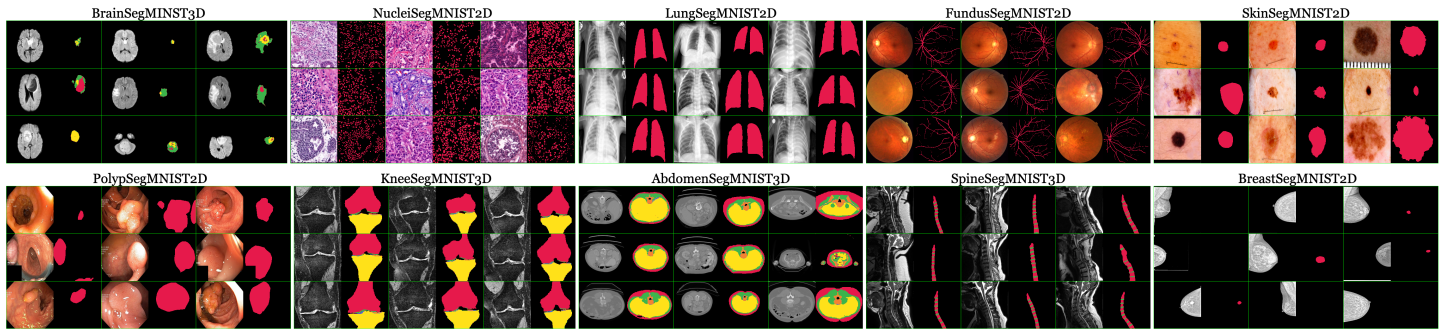


Figure 1: Representative samples from datasets included in MedSegMNIST plotted in a 3×3 grid. For volumetric datasets, a representative slice from the corresponding volume is shown with segmentation annotations to the left.

multi-class volumetric segmentation problems.

The four volumetric datasets consist of abdominal organ segmentation from CT (Rister et al., 2020), brain tumour segmentation from MRI (Adewole et al., 2025), cervical spine structure segmentation from MRI (Zhou et al., 2025), and knee cartilage segmentation from MRI (Ambellan et al., 2019). AbdomenSegMNIST3D is derived from CT-ORG (Rister et al., 2020) and contains multi-organ abdominal CT volumes with five semantic classes. BrainSegMNIST3D is based on BraTS-Africa (Adewole et al., 2025) and provides three-class brain tumour segmentation masks from T2-FLAIR MRI volumes. SpineSegMNIST3D is derived from DukeCSpineSeg (Zhou et al., 2025) and focuses on cervical spine structure segmentation, while KneeSegMNIST3D is constructed from the OAIZIB-CM knee MRI dataset (Ambellan et al., 2019) and contains multiple cartilage compartments relevant to osteoarthritis research. All volumetric datasets are distributed at several standardised resolutions together with native-resolution variants.

The six 2D datasets cover a diverse range of clinical imaging applications. LungSegMNIST2D combines chest radiographs from multiple public sources to provide large-scale lung field segmentation (Danilov et al. (2022)). BreastSegMNIST2D contains mammography images with lesion annotations for breast tumour segmentation (Oza et al., 2024). FundusSegMNIST2D is derived from the FIVES dataset (Jin et al., 2022) and focuses on retinal vessel segmentation from colour fundus photographs. NucleiSegMNIST2D combines histopathology datasets (Kumar et al., 2019; Samet et al., 2025) for nuclei segmentation in H&E-stained tissue images. PolypSegMNIST2D is based on Kvasir-SEG (Jha et al., 2019) and provides binary segmentation masks for colorectal polyps in endoscopic imagery. SkinSegMNIST2D is derived from the ISIC 2018 challenge (Codella et al., 2019) and contains dermoscopic lesion boundary annotations.

These datasets encompass grayscale and RGB imaging modalities, varying image resolutions, and both binary and multi-class segmentation settings. The compressed NPZ files are to be hosted on zenodo, for

A summary of the datasets included in MedSegMNIST is provided in Table 2.

6. Baseline Benchmarking

To demonstrate the usability of the MedSegMNIST benchmarking pipeline, a baseline U-Net model (Ronneberger et al., 2015) is trained on each dataset using the predefined train-test partitions. Experiments were intentionally lightweight, consisting of five training epochs with no dataset-specific hyperparameter tuning. The objective is to validate the end-to-end functionality of the MedSegMNIST training and evaluation workflow.

Table 3 summarizes the resulting segmentation performance. Substantial variability is observed across datasets. Simpler anatomical segmentation tasks such as LungSegMNIST2D and KneeSegMNIST3D achieve high Dice scores after only a small number of training epochs, whereas more challenging tasks such as BrainSegMNIST3D and BreastSegMNIST2D remain difficult under the same training budget. These results highlight the diversity of the included tasks and illustrate the utility of MedSegMNIST as a common evaluation environment for learning, prototyping, and benchmarking segmentation methods across different imaging modalities, anatomical structures, and problem complexities.

7. MedSegMNIST Functionalities

The MedSegMNIST API (1) exposes the following core functionalities:

Dataset access Per-dataset classes (BrainSegMNIST3D (split, size, root, transform)) returning (image, mask) pairs; a get_data() method returning the full raw array without the channel dimension for non-PyTorch workflows.

Cross-validation get_fold(k) returns (train_subset, test_subset) as PyTorch Subset objects for any of the five pre-defined folds.

Table 2: MedSegMNIST Dataset Summary. Dim. = dimensionality; N = number of subjects/images. All datasets come with native resolution.

Flag	Class	Modality	Dim.	Labels	N	Sizes (px)			Source
brain3d	BrainSegMNIST3D	MRI	3D	3	146	96	128	224	(Adewole et al., 2025)
abdomen3d	AbdomenSegMNIST3D	CT	3D	5	140	64	96	128 192	(Rister et al., 2020)
spine3d	SpineSegMNIST3D	MR	3D	2	1250	64	96	128 192	(Zhou et al., 2025)
knee3d	KneeSegMNIST3D	MR	3D	5	507	64	96	128 192	(Ambellan et al., 2019)
lung2d	LungSegMNIST2D	X-ray	2D	1	6810	128	256	512	(Danilov et al., 2022)
nuclei2d	NucleiSegMNIST2D	Pathology	2D	1	151		256	512	(Kumar et al., 2019; Samet et al., 2025)
polyp2d	PolypSegMNIST2D	Endoscopy	2D	1	1000	128	256	512	(Jha et al., 2019)
breast2d	BreastSegMNIST	Mammography	2D	1	511	128	256	512	(Oza et al., 2024)
fives2d	FundusSegMNIST2D	Fundus	2D	1	800	256	512	1024	(Jin et al., 2022)
derm2d	SkinSegMNIST2D	Dermoscopy	2D	1	2594	128	256	512	(Codella et al., 2019)

Table 3: Reference U-Net results obtained using the MedSegMNIST benchmarking pipeline. Results are reported on the predefined test partitions.

Dataset	Res.	Dice $\uparrow$	IoU $\uparrow$	HD95 $\downarrow$	Time (s)
NucleiSegMNIST2D	256	0.6990	0.5429	9.67	10
PolypSegMNIST2D	256	0.4288	0.3021	98.33	45
FundusSegMNIST2D	256	0.7588	0.6184	12.46	40
SkinSegMNIST2D	256	0.7782	0.6810	34.91	111
LungSegMNIST2D	256	0.9525	0.9104	10.10	289
BreastSegMNIST2D	256	0.1715	0.1111	95.23	29
AbdomenSegMNIST3D	192	0.7738	0.6509	16.47	589
BrainSegMNIST3D	224	0.1894	0.1311	37.71	325
SpineSegMNIST3D	128	0.4725	0.3198	12.85	714
KneeSegMNIST3D	128	0.8907	0.8116	1.54	408

Physical metadata `get_voxel_spacing()` returns the millimetre spacing associated with the active size variant, required for physically calibrated surface-distance metrics.

Registry `list_datasets()` and `info(flag)` allow programmatic discovery of available datasets and their metadata (modality, classes, available sizes, source citation) without instantiating a dataset.

Visualisation `plot_sample`, `plot_overlay`, `plot_grid` utilities, with automatic handling of greyscale vs. RGB display, label-aware colour-mapping, and slice-selection defaults appropriate to each 3D dataset’s anatomical viewing convention.

Benchmarking pipeline A `train.py` / `eval.py` command-line pair that accepts any user-supplied PyTorch segmentation model and a `flag` argument selecting any registered dataset, computing the full evaluation metric suite (Dice, HD95, ASSD, RVE, NSD) without requiring the user to implement metric computation themselves.

Listing 1: MedSegMNIST API Usage Example

```
# $ pip install medsegmnist[torch]

import medsegmnist as msm
from monai.models import UNet

train_set = msm.PolypSegMNIST2D(split="train",
                                size=256)

msm.plot_sample(train_set[0])
model = UNet(in_channels=3, out_channels=1)

results = msm.benchmark(model=model,
                        flag="polyp2d", folds=5)

print(results)
```

8. Conclusion

This work presented MedSegMNIST, an open-source benchmark and software package for medical image segmentation that aims to improve accessibility, reproducibility, and ease of experimentation across diverse segmentation tasks. The resource brings together ten curated datasets spanning seven imaging modalities, multiple anatomical regions, and both 2D and 3D segmentation settings under a common interface for data access, visualization, training, and evaluation.

By providing standardized preprocessing, predefined cross-validation splits, multiple resolution variants, and integrated benchmarking utilities, MedSegMNIST reduces the engineering overhead often associated with dataset preparation and model comparison. The inclusion of both native-resolution and resized dataset variants further enables investigations across different computational and methodological settings.

MedSegMNIST is envisioned as a resource for both education and research, supporting exploratory learning, rapid prototyping, and reproducible benchmarking. Future work will focus on expanding dataset coverage, incorporating additional segmentation tasks and modalities, and extending benchmarking capabilities to support emerging foundation and multimodal models for medical image analysis.

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